

Mitutoyo 52-Bit Data Stream: Complete Guide

52-bit Frame Breakdown

Mitutoyo 52-bit Frame (13 nibbles, 4 bits each)

N0	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12
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Wiring Connections

Caliper Pin	Signal	Arduino Pin (via level shifting)
1	GND	GND
2	DATA	D4
3	CLOCK	D2 (INT0)
4	REQ	D3 (optional)

Canonical Arduino Code

```
/*
  Mitutoyo / Digimatic 52-bit reader
  - Captures 52 bits LSB-first on CLK falling edge
  - Prints raw 13 nibbles and a best-effort decoded value
  - Works with SPC Digimatic calipers, micrometers, indicators
*/

#define CLK_PIN 2 // INT0
#define DATA_PIN 4
#define REQ_PIN 3

const bool USE_REQ = true;
const unsigned long REQ_LOW_MS = 120;
const unsigned long FRAME_TIMEOUT_MS = 200;

volatile uint64_t frameBits = 0;
volatile uint8_t bitCount = 0;
volatile bool frameReady = false;

void isrClockFalling() {
  uint8_t bit = digitalRead(DATA_PIN) ? 1 : 0;
  if (bitCount < 64) {
    frameBits |= (uint64_t)bit << bitCount;
    bitCount++;
    if (bitCount == 52) {
      frameReady = true;
      detachInterrupt(digitalPinToInterrupt(CLK_PIN));
    }
  }
}

void startCapture() {
```

```

    noInterrupts();
    frameBits = 0;
    bitCount = 0;
    frameReady = false;
    interrupts();
    attachInterrupt(digitalPinToInterrupt(CLK_PIN), isrClockFalling, FALLING);
}

void sendRequestPulseIfEnabled() {
    if (!USE_REQ) return;
    digitalWrite(REQ_PIN, LOW);
    delay(REQ_LOW_MS);
    digitalWrite(REQ_PIN, HIGH);
}

void setup() {
    Serial.begin(115200);
    pinMode(CLK_PIN, INPUT_PULLUP);
    pinMode(DATA_PIN, INPUT_PULLUP);
    if (USE_REQ) {
        pinMode(REQ_PIN, OUTPUT);
        digitalWrite(REQ_PIN, HIGH);
    }
    Serial.println("Mitutoyo 52-bit reader ready.");
}

bool captureOneFrame(uint64_t &outBits) {
    startCapture();
    sendRequestPulseIfEnabled();
    unsigned long t0 = millis();
    while (!frameReady && (millis() - t0) < FRAME_TIMEOUT_MS) {}
    detachInterrupt(digitalPinToInterrupt(CLK_PIN));
    if (!frameReady) return false;
    noInterrupts();
    outBits = frameBits;
    interrupts();
    return true;
}

void loop() {
    uint64_t bits;
    if (captureOneFrame(bits)) {
        Serial.print("Raw frame: ");
        for (int i = 0; i < 13; i++) {
            Serial.print((bits >> (i*4)) & 0xF, HEX);
            Serial.print(" ");
        }
        Serial.println();
    }
    delay(50);
}

```

This document combines the history, data stream documentation, wiring, and a complete working Arduino sketch into a single reference for interfacing with Mitutoyo Digimatic tools.